

ADM Training Case Study and Assignments

1. Meridian Hospitality Group - Master Narrative

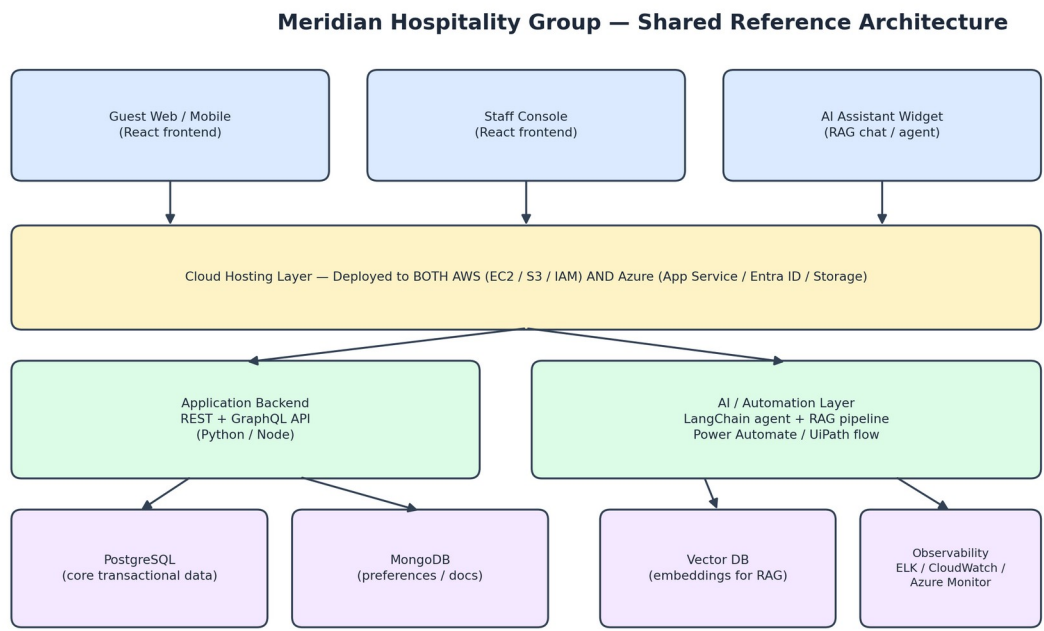
Meridian Hospitality Group is a fictional hospitality holding company operating five distinct business units. All five units run on a shared technology platform managed by a small central engineering team - the role each cohort team plays in this bootcamp. The group's mandate for this engineering effort: modernize each unit's guest and operations experience with a secure, cloud-hosted application layer, then extend it with AI features and one automated workflow, and finally take it to production with proper monitoring.

Team	Brand	Business Unit	Cloud Deployment
Team 1	The Aldwyn House	Boutique / Independent Hotel	AWS + Azure
Team 2	Meridian Resorts & Spa	Resort & Spa Chain (multi-property)	AWS + Azure
Team 3	Meridian Kitchens Collective	Restaurant / F&B Group	AWS + Azure
Team 4	Meridian Stays	Vacation Rental / Property Management Company	AWS + Azure
Team 5	Meridian Residences	Extended-Stay / Serviced Apartments	AWS + Azure

2. Shared Starter Baseline (given to all teams)

This is the common foundation every team starts from on Day 10. It represents the assumed output of Sprint 1: an already-agreed data model, API contract, architecture, and a working Dockerfile + docker-compose.yml for the application itself. Teams may extend this baseline with the vertical-specific entities listed in their brief (Section 4), but should not need to redesign it or write container packaging from scratch.

2.1 Reference Architecture



2.2 Core Data Model (shared across all teams)

Entity	Key Fields	Notes
Guest	id, name, email, phone, loyalty_tier, created_at	Core guest profile shared across all touchpoints
Property	id, name, brand, address, timezone	A physical hotel / restaurant / unit / building
Reservation	id, guest_id, property_id, check_in, check_out, status, rate_plan_id	Core booking record for a property stay
RatePlan	id, property_id, name, nightly_rate, cancellation_policy	Pricing rules applied to a reservation
Folio	id, reservation_id, line_items[], balance, status	Guest bill / invoice for a stay
Order	id, property_id, guest_id (nullable), items[], total, placed_at	An F&B order placed at a property outlet

2.3 Core API Contract (shared across all teams)

Endpoint	Purpose
GET /api/v1/reservations	List / filter reservations by property, date range, status
GET /api/v1/reservations/{id}	Retrieve a single reservation with guest + folio detail
POST /api/v1/reservations	Create a reservation (used by booking flows)
GET /api/v1/guests/{id}	Retrieve guest profile + preference document (Mongo)
GET /api/v1/folios/{id}	Retrieve a folio and its line items
GET /api/v1/availability	Check room / slot availability for a property + date range

2.4 Stack & Deployment Notes

- Backend: Python or Node REST + GraphQL API; Dockerfile and docker-compose.yml provided in the starter pack - teams extend the application code, not the container setup.
- Databases: PostgreSQL for transactional data (Guest, Reservation, Folio, Order); MongoDB for semi-structured guest preference/document data.
- Frontend: React, using hooks and Axios against the REST API.
- Cloud: every team deploys to both AWS (single EC2 instance running the provided Docker image) and Azure (App Service for Containers, using staging/production deployment slots), using the IAM / Azure AD patterns from Week 3.
- Environments: Dev, QA, and UAT run as automated gates inside the CI/CD pipeline (build, lint, test) with no separate live infrastructure; Staging and Production are the only live environments - Staging/Production on Azure are the App Service deployment slots, and on AWS are handled by redeploying to the same EC2 instance.

3. Sprint Structure (applies to every team)

Sprint 2 - Cloud, Secure Integration & UI (Days 10–14)

- Use an AI coding assistant (Cursor/Codex) within compliance guardrails - no guest PII in prompts - to scaffold one feature.
- Apply prompt engineering to prototype the AI feature planned for Sprint 3.
- Build a React frontend for the one core workflow specified in your team brief, consuming the shared REST API.
- Deploy the provided Docker image to a single EC2 instance on AWS and to an Azure App Service container on Azure - your team's first live deployment, done manually.

Deliverable:

AI-assisted code module + working frontend for your core workflow, live on both AWS and Azure.

Sprint 3 - AI Feature & Automation Build (Days 15–19)

- Build the RAG-based assistant specified in your team brief using a vector database over your unit's reference documents.
- Implement the AI agent specified in your team brief, including a clear stopping condition.
- Automate the one supporting workflow specified in your team brief using Power Automate or UiPath.

Deliverable:

Working RAG/agent feature integrated into your application + one automated workflow.

Final Sprint - Production Readiness (Days 20–21)

- Automate your Sprint 2 deployment into a CI/CD pipeline (GitHub Actions or Jenkins) that promotes a build through Dev → QA → UAT (automated checks) to Staging → Production (Azure slot swap; AWS redeploy to the EC2 instance).
- Set up logging, monitoring dashboards, and at least one alert (ELK, CloudWatch, or Azure Monitor).
- Triage the simulated incident specified in your team brief using Jira Service Management, following incident do's/don'ts covered on Day 20.
- Present a live demo of the full application (core workflow + AI feature + automation + monitoring dashboard) to the assessment panel.

Deliverable:

Fully deployed application + monitoring setup + live demo + final assessment.

4. Team Briefs

Team 1: The Aldwyn House

Boutique / Independent Hotel

A 40-room independent luxury boutique hotel built on high-touch, personalized guest experience.

Business context

The Aldwyn House competes on personalization, not scale. Guests expect staff to remember their preferences, anticipate requests, and respond to questions instantly - but the hotel's small team can't manually track every preference across stays. Leadership wants an AI-assisted guest experience layer on top of the existing reservation and folio data.

Unique data entities (extend the shared baseline)

- GuestPreferenceProfile (dietary, room, past requests)
- ConciergeRequest (request type, status, assigned staff)
- Amenity (spa, dining, local experiences catalogue)

Sprint 2 - core workflow to build

A staff-facing “Guest 360” dashboard (React) that pulls reservation + preference data via the REST API, showing upcoming arrivals and flagged preferences.

Sprint 2 - AI-assisted build note

Use an AI coding assistant to scaffold the Guest 360 dashboard components and a preference-matching function, with guardrails preventing guest PII from being sent to the LLM in prompts.

Sprint 3 - RAG assistant

An AI concierge chat widget that answers guest questions (“What time does the spa close?”, “Is the restaurant pet-friendly?”) by retrieving answers from the hotel's policy and amenities documents.

Sprint 3 - AI agent task

An agent that handles routine concierge requests end-to-end: interpret a guest's free-text request (e.g., “late checkout”, “extra pillows”), check eligibility against the reservation record, and create/update a ConciergeRequest.

Sprint 3 - automated workflow

Power Automate flow: on new guest check-in, automatically notify housekeeping of any flagged preferences and log a welcome-amenity task.

Final Sprint - simulated incident

Simulated incident: the preference sync job between the reservation system and the app database silently fails for one night - team triages via Jira Service Management and documents root cause + fix.

Team 2: Meridian Resorts & Spa

Resort & Spa Chain (multi-property)

A 6-property beach & mountain resort chain bundling rooms, spa, and multiple F&B outlets under one loyalty umbrella.

Business context

Meridian Resorts & Spa runs multiple properties, each with its own restaurants and spa calendars, but guests book and expect a single unified experience across whichever property they're at. Operations wants one dashboard that spans properties and a way to upsell spa/dining during a guest's stay.

Unique data entities (extend the shared baseline)

- SpaAppointment (property, therapist, service, time slot)
- MultiPropertyInventory (room + spa slot availability by property)
- UpsellOffer (targeted offer per guest segment)

Sprint 2 - core workflow to build

A cross-property operations dashboard (React) showing today's arrivals, spa bookings, and F&B covers across all 6 properties, consuming the REST API.

Sprint 2 - AI-assisted build note

Use an AI coding assistant to scaffold a multi-property availability aggregation component.

Sprint 3 - RAG assistant

A guest-facing AI assistant answering questions about spa treatments and dining options across properties, grounded in each property's service menu and hours documents.

Sprint 3 - AI agent task

An agent that recommends and books a spa slot for a guest based on their reservation dates, stated preferences, and real-time availability, stopping to ask for confirmation before finalizing.

Sprint 3 - automated workflow

UiPath workflow: automatically reconcile nightly F&B covers against expected occupancy and flag properties with unusual variance for manager review.

Final Sprint - simulated incident

Simulated incident: a spike in spa booking failures during a promotional weekend - team triages, identifies the bottleneck, and documents an incident report.

Team 3: Meridian Kitchens Collective

Restaurant / F&B Group

A group of 5 branded restaurant concepts sharing a central kitchen, loyalty program, and inventory system.

Business context

Meridian Kitchens Collective wants to modernize how its restaurants manage inventory, loyalty points, and menu Q&A, all built on top of point-of-sale (POS) data. Frequent stockouts of popular items and slow allergen/ingredient lookups are hurting service speed.

Unique data entities (extend the shared baseline)

- MenuItem (ingredients, allergens, price, outlet)
- LoyaltyAccount (points balance, tier, redemption history)
- InventoryStock (ingredient, outlet, quantity, reorder threshold)

Sprint 2 - core workflow to build

A manager dashboard (React) showing live inventory levels per outlet and loyalty program performance, consuming the REST API layered over POS data.

Sprint 2 - AI-assisted build note

Use an AI coding assistant to scaffold an inventory low-stock alert component and a loyalty-tier calculation function.

Sprint 3 - RAG assistant

A guest-facing AI assistant answering menu and allergen questions (“Which dishes are nut-free?”) grounded in the current menu and ingredient documents.

Sprint 3 - AI agent task

An agent that monitors InventoryStock and, when an ingredient crosses its reorder threshold, drafts a reorder request with quantity and supplier for a manager to approve.

Sprint 3 - automated workflow

Power Automate flow: when a loyalty member crosses a points threshold, automatically send a personalized reward notification and log the redemption offer.

Final Sprint - simulated incident

Simulated incident: the POS-to-inventory sync double-counts stock after a network blip - team triages via Jira Service Management and documents the fix.

Team 4: Meridian Stays

Vacation Rental / Property Management Company

A property management company operating 150+ owner-listed vacation rental units across multiple cities.

Business context

Meridian Stays manages listings on behalf of individual property owners, coordinates cleaning/turnover between stays, and owes each owner clear performance reporting. Owners currently get manual monthly reports; the business wants near-real-time visibility and faster guest-question handling.

Unique data entities (extend the shared baseline)

- OwnerAccount (owner profile, payout terms, properties owned)
- UnitListing (property, amenities, nightly rate, calendar)
- CleaningTask (unit, turnover window, assigned vendor, status)

Sprint 2 - core workflow to build

An owner-facing performance dashboard (React) showing occupancy, revenue, and upcoming bookings per unit, plus a staff view of the cleaning/turnover schedule, both via the REST API.

Sprint 2 - AI-assisted build note

Use an AI coding assistant to scaffold the owner dashboard's revenue chart component and the turnover-scheduling logic.

Sprint 3 - RAG assistant

A guest-facing AI assistant answering listing-specific questions (Wi-Fi password policy, check-in instructions, house rules) grounded in each unit's listing documents.

Sprint 3 - AI agent task

An agent that, on a new booking, automatically schedules the cleaning/turnover task with the right vendor and time window, checking for conflicts before confirming.

Sprint 3 - automated workflow

Power Automate flow: after each stay, automatically send the guest a review request and generate/send the monthly owner performance summary.

Final Sprint - simulated incident

Simulated incident: double-booking appears on one unit's calendar due to a sync delay between the booking API and the calendar service - team triages and documents resolution.

Team 5: Meridian Residences

Extended-Stay / Serviced Apartments

A serviced-apartment brand for 30+ day corporate and relocation stays, blending hotel-style service with apartment-style billing.

Business context

Meridian Residences bills guests on lease-like recurring cycles (weekly/monthly) rather than a single folio, and handles a steady stream of maintenance requests across long-stay units. Finance needs cleaner recurring billing, and guests want faster maintenance turnaround.

Unique data entities (extend the shared baseline)

- LeaseAgreement (unit, guest, term, monthly rate, renewal date)
- MaintenanceTicket (unit, issue type, priority, status)
- RecurringInvoice (lease, billing period, amount, payment status)

Sprint 2 - core workflow to build

A resident portal (React) where guests view their lease, upcoming invoices, and can submit maintenance requests, backed by the REST API.

Sprint 2 - AI-assisted build note

Use an AI coding assistant to scaffold the recurring-invoice generation component and the maintenance-ticket submission form.

Sprint 3 - RAG assistant

A resident-facing AI assistant answering lease and policy questions (“What’s the early termination policy?”, “Is parking included?”) grounded in the lease-terms and building-policy documents.

Sprint 3 - AI agent task

An agent that triages incoming maintenance requests by priority/type and assigns them to the right vendor queue, escalating urgent issues (e.g., no heat, water leak) automatically.

Sprint 3 - automated workflow

UiPath workflow: on the 1st of each month, automatically generate and send RecurringInvoice records for all active leases and flag any overdue payments from the prior cycle.

Final Sprint - simulated incident

Simulated incident: the monthly invoice-generation job fails for a subset of leases - team triages via Jira Service Management, identifies affected leases, and documents the fix.

5. Individual Assignments (Days 10–21)

These run alongside the team capstone and are submitted by each learner individually. The summary below shows each day's individual assignment; submit each one by the due date shared by your facilitator.

Day	Topic	Individual Assignment
10	Cloud Fundamentals – AWS	Launch/configure an EC2 instance; create an S3 bucket with a lifecycle policy; write a least-privilege IAM policy for one role. Submit config + policy JSON.
11	Cloud Fundamentals – Azure & RBAC	Deploy a sample app to an Azure App Service deployment slot; register it in Azure AD; write a short AWS-IAM-vs-Azure-AD RBAC comparison note.
12	Prompt Engineering	Write 5 prompts for a hospitality task (system/user/context framing, few-shot, output schema). Submit prompt log + model outputs.
13	LLMs & AI Compliance	Compare GPT vs. Claude on one coding task; write a short data-privacy checklist for using third-party LLMs with guest PII.
14	React Basics	Build a small React component consuming a REST endpoint (loading/error states) using an AI coding assistant. Submit component + prompt transcript.
15	LangChain	Build a LangChain chain with a prompt template and conversation memory for a 2-step hospitality task.
16	RAG / Vector Databases	Build embeddings + similarity search over a small hospitality document set using a vector database.
17	Agentic Workflows	Design an agent decision loop with a clearly defined stopping condition for one task.
18	Power App Platforms	Build one Canvas app screen + one Power Automate flow + one Power BI report against sample data.
19	UiPath	Build a UiPath workflow with reliable selectors for one repetitive task; simulate an Orchestrator queue.
20	CI/CD, Monitoring & Incident Triage	Configure one CI/CD pipeline stage; set up one CloudWatch/Azure Monitor alert; triage a simulated Jira Service Management ticket.
21	Final Demo	Team demo to the panel + individual peer-contribution submission + final assessment.